# 🏦 AI Banking Assistant

*Intelligent Voice-Enabled Banking System with Secure Authentication*

A Full-Stack AI Application for Modern Banking Services

Technical Documentation & Project Overview

By Micheal Olaniyi [AI/Machine Learning Engineer]

## Executive Summary

The AI Banking Assistant is a sophisticated, production-ready application that combines natural language processing, voice recognition, and secure authentication to deliver a seamless banking experience. Built with OpenAI's GPT-4o-mini, the system handles customer authentication, account management, transaction queries,loan eligibility checks, and professional advice and information for each aunthenticated customers through both voice and text interfaces.

| 700 | 1000+ | 6 | 2FA |
|---|---|---|---|
| Mock Customers | Transactions | Banking Tools | Authentication |

### 🎯 Project Goals

- ✓ Demonstrate practical AI implementation in financial services
- ✓ Create secure, user-friendly banking interface
- ✓ Enable voice-first interactions for accessibility

- ✓ Showcase full-stack development capabilities
- ✓ Build production-ready, scalable architecture

# Key Features

## 🔐 Secure Two-Factor Authentication

- ✓ **CX Code Verification:** Unique 8-character customer code (format: A1234XYZ - 1 letter + 4 numbers + 3 letters)
- ✓ **Security Questions:** Personalized security question verification with case-insensitive matching
- ✓ **Session Management:** Secure, isolated customer sessions with automatic timeout
- ✓ **Access Control:** Prevents unauthorized access to other customer data through strict ID filtering
- ✓ **Authentication Flow:** Multi-step verification process ensuring legitimate user access

## 🤖 AI-Powered Banking Operations

- ✓ **Natural Conversation:** Chat naturally using everyday language about banking needs
- ✓ **Account Balance:** Real-time balance checking for checking and savings accounts
- ✓ **Transaction History:** View recent transactions with detailed information
- ✓ **Transaction Search:** Filter transactions by amount, date range, and type
- ✓ **Loan Eligibility:** Instant pre-qualification assessment based on credit score and income
- ✓ **Loan Information:** Detailed information about available loan products and terms

✓ **Smart Context:** AI maintains conversation context across multiple queries

## 🎤 Voice Interaction

✓ **Speech-to-Text:** OpenAI Whisper API for accurate voice transcription in English

✓ **Text-to-Speech:** Natural voice responses using OpenAI TTS-1 with multiple voice options

✓ **Multi-modal Interface:** Switch seamlessly between voice and text input

✓ **Accessibility:** Enhanced accessibility for users preferring voice interaction

✓ **Real-time Processing:** Sub-2-second voice processing for natural conversation flow

# Technical Architecture

## Technology Stack

### 🧠 AI & NLP

**OpenAI GPT-4o-mini**
• Function calling for tool integration
• Context-aware conversations
• Natural language understanding
• Intent recognition

### 🎤 Voice Processing

**Whisper API**
• Speech recognition
• Multi-language support

**TTS-1 API**
• Voice synthesis
• Multiple voice options

### 💻 Backend

**Python 3.8+**
• SQLite Database
• Custom tool integration
• RESTful architecture
• Modular design pattern

### 🖥️ Frontend

**Gradio Framework**
• Responsive web interface
• Real-time audio processing
• Dual input modes
• Mobile-friendly design

## Database Schema

The system uses a relational SQLite database with five interconnected tables ensuring data integrity through foreign key constraints:

| Table | Purpose | Key Fields | Relationships |
|---|---|---|---|
| **customers** | Core customer information | customer_id, name, email, credit_score, annual_income | 1:1 with auth, 1:many with accounts |
| **customer_auth** | Authentication credentials | cx_code, security_question, security_answer | 1:1 with customers |

| Table | Purpose | Key Fields | Relationships |
|---|---|---|---|
| **accounts** | Banking accounts | account_id, account_type, balance, status | Many:1 with customers, 1:many with transactions |
| **transactions** | Financial transactions | transaction_id, amount, type, date, description | Many:1 with accounts |
| **cards** | Debit/Credit cards | card_id, card_number, card_type, expiry_date | Many:1 with accounts |

**Data Integrity Features:**

- Foreign key constraints ensure referential integrity across all tables
- Automated verification scripts check for orphaned records
- Transaction timestamps provide complete audit trails
- Cascading operations maintain data consistency
- Parameterized queries prevent SQL injection attacks

# System Workflow

## 📊 Complete Process Flow Diagram

**START: User Opens App**

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### 🔐 AUTHENTICATION PATH

**1. User Input (Voice/Text) → 2. Speech-to-Text (Whisper) → 3. Authentication Check**

❌ **Not Authenticated:** Ask CX Code → Verify in DB → Ask Security Question → Verify Answer → Create Session

✅ **Already Authenticated:** Proceed to Banking Operations

↓

### 🏦 BANKING OPERATIONS PATH

**4. LLM Processing (GPT-4o-mini)**
Analyzes intent → Determines required tool

**→ Check Balance Tool**

**→ Get Transactions Tool**

**→ Check Loan Eligibility Tool**

→ **Search Transactions Tool**

**5. Database Query** (SQLite)
Secure parameterized query with customer_id filter

**6. Tool Returns Results** → Format data with currency & dates

↓

**RESPONSE GENERATION PATH**

**7. LLM Generates Natural Response**
Incorporates data into conversational format

**8. Text-to-Speech (TTS-1 API)**
Converts response to natural voice

**Display Text** **Play Audio**

**9. Update Chat History**
Maintain context for next query

**Wait for Next Input**

*Complete end-to-end process flow from user input through authentication, AI processing, database queries, to final voice/text response*

## Detailed Authentication Flow

### Step 1: Initial Contact

User initiates conversation via voice or text interface. System welcomes user and begins authentication process.

#### Step 2: CX Code Request

System prompts for 8-character CX code (format: letter + 4 digits + 3 letters, e.g., A1234XYZ).

#### Step 3: Database Verification

System validates CX code against `customer_auth` table using case-insensitive comparison.

#### Step 4: Security Question Retrieval

Upon successful CX code validation, system retrieves customer's personalized security question.

#### Step 5: Security Question Presentation

System asks the retrieved security question and awaits user response.

#### Step 6: Answer Validation

System performs case-insensitive comparison with stored answer, trimming whitespace for accuracy.

#### Step 7: Session Creation

Upon successful authentication, secure session is established with customer_id linkage and unique session identifier.

#### Step 8: Welcome & Account Summary

System greets customer by name, displays comprehensive account summary including all accounts, balances, and available services.

## Banking Operations Flow

#### Step 1: User Request Reception

Customer asks banking question through voice (transcribed via Whisper) or text input.

### Step 2: Intent Recognition

GPT-4o-mini analyzes request, identifies intent, and determines required banking operation.

### Step 3: Tool Selection

AI model selects appropriate banking tool (balance check, transaction history, loan eligibility, etc.) via function calling.

### Step 4: Parameter Extraction

System extracts necessary parameters from user request (account type, date ranges, amounts, etc.).

### Step 5: Security Validation

System verifies session is authenticated and customer_id matches for secure data access.

### Step 6: Database Query Execution

Tool executes secure parameterized SQL query with customer_id filter to prevent unauthorized access.

### Step 7: Data Processing & Formatting

Results are formatted with currency symbols, dates, and structured for optimal AI interpretation.

### Step 8: Response Generation

GPT-4o-mini crafts natural, conversational response incorporating retrieved data and context.

### Step 9: Output Delivery

Response is displayed as text in chat interface and simultaneously converted to speech via TTS-1 API.

### Step 10: History Update

> Conversation history is updated to maintain context for subsequent queries.

## Banking Tools & Functions

The system implements 8 specialized banking tools integrated via OpenAI's function calling mechanism:

| Tool Name | Function | Parameters | Output |
|---|---|---|---|
| `verify_cx_code` | Validates customer CX code | cx_code (string) | customer_id, security_question, validation status |
| `verify_security_answer` | Validates security question response | customer_id, security_answer | authentication status, customer details, account summary |
| `get_customer_welcome_info` | Retrieves comprehensive customer data | customer_id | full profile, all accounts, total balance, recent activity |
| `check_balance` | Retrieves specific account balance | customer_id, account_type | current balance, account status, account details |
| `get_all_accounts` | Lists all customer accounts | customer_id | account list with types, balances, and total |
| `get_recent_transactions` | Fetches transaction history | customer_id, account_type, limit | transaction list with dates, amounts, descriptions |
| `search_transactions` | Filters transactions by | customer_id, min/max | filtered transactions, |

| Tool Name | Function | Parameters | Output |
|---|---|---|---|
| | criteria | amount, days | spending totals, statistics |
| `check_loan_eligibility` | Assesses loan qualification | customer_id, loan_amount, loan_type | eligibility status, estimated rate, monthly payment |

## Tool Implementation Details

**Function Calling Architecture**

Each tool is registered with OpenAI's function calling system through JSON schema definitions that specify:

- **Name:** Unique identifier for the function
- **Description:** Clear explanation of what the function does (helps AI decide when to use it)
- **Parameters:** JSON schema defining required and optional parameters with types and descriptions
- **Return Type:** Expected output structure

The AI model analyzes user requests and autonomously decides which tools to call, extracting necessary parameters from natural language input.

# Security & Privacy

### 🔒 Security Measures Implemented

- ✓ **Two-Factor Authentication:** CX code + security question provides robust identity verification
- ✓ **Session Isolation:** Each user session is completely isolated with unique identifiers preventing cross-contamination
- ✓ **Data Access Control:** All database queries filtered by authenticated customer_id ensuring users can only access their own data
- ✓ **SQL Injection Prevention:** Parameterized queries used throughout eliminating SQL injection vulnerabilities
- ✓ **API Key Security:** Sensitive credentials stored in environment variables, never in code
- ✓ **Input Validation:** All user inputs sanitized and validated before processing
- ✓ **Session Timeout:** Automatic session expiration after inactivity period
- ✓ **HTTPS Enforcement:** Secure communication via encrypted connections for voice features

**⚠️ Demo vs Production Considerations:**

This is a demonstration application using mock data. For production deployment, additional security measures would include:

- **Password Hashing:** Bcrypt or Argon2 for secure password storage
- **JWT Tokens:** Token-based authentication with refresh mechanisms
- **Rate Limiting:** Prevent brute force attacks and abuse
- **Audit Logging:** Comprehensive logging of all operations for security audits
- **Database Encryption:** Encryption at rest for sensitive data
- **DDoS Protection:** Cloudflare or similar service for protection against attacks
- **Compliance:** PCI DSS, GDPR, and regional banking regulations adherence

- **Penetration Testing:** Regular security assessments

# Performance & Scalability

## Current Performance Metrics

✓ **Database Operations:** 700 customers, 1000+ transactions with sub-second query times

✓ **API Response Time:** < 2 seconds for typical banking queries including AI processing

✓ **Voice Processing:** ~1-2 seconds for speech-to-text transcription, ~1 second for text-to-speech synthesis

✓ **Concurrent Users:** Supports multiple simultaneous sessions with isolated state management

✓ **Scalability:** Horizontally scalable architecture ready for cloud deployment

## Optimization Strategies Implemented

### Database Optimization

- Indexed primary and foreign keys
- Efficient query design
- Connection pooling ready

### Caching Strategy

- Session state caching
- Customer data caching
- Redis integration ready

### API Efficiency

- Batched operations
- Async processing capable
- Minimal API calls

### Load Balancing

- Stateless architecture
- Horizontal scaling ready
- Multi-instance capable

# Development Process

## Project Structure

```
ai-banking-assistant/
├── app.py                          # Gradio interface & main
├── llm_handler.py                 # OpenAI integration & tool
├── audio_processor.py             # Voice processing (STT/TTS
├── .env                           # Environment variables (API
├── requirements.txt               # Python dependencies
├── README.md                      # Project documentation
└── tools/                         # Banking function modules
    ├── __init__.py
    ├── auth.py                 # Authentication functions (C
    ├── balance.py              # Account balance checking to
    ├── transactions.py         # Transaction history & searc
    └── loans.py                # Loan eligibility & informat
└── banking.db/                    # SQLite database (auto-gen
    ├── test_database.py              # Data integrity verific
    ├── db_setup.py                   # Database schema creati
    └── mock_data.py
```

## Development Phases

### Phase 1: Foundation & Planning (Week 1)

Database design, schema creation, relationships mapping, and data modeling. Established core architecture and technology stack decisions.

### Phase 2: Authentication System (Week 1-2)

Implementation of two-factor authentication with CX code generation, security question system, and session management infrastructure.

### Phase 3: Banking Tools Development (Week 2)

Created 6 core banking function tools: balance checking, account management, transaction queries, search functionality, and loan services.

### Phase 4: AI Integration (Week 2-3)

OpenAI GPT-4o-mini integration with function calling, prompt engineering for banking context, and tool orchestration logic.

### Phase 5: Voice Features (Week 3)

Speech-to-text implementation using Whisper API and text-to-speech using TTS-1 API with multi-voice support.

### Phase 6: User Interface (Week 3)

Gradio interface development with dual text/voice tabs, responsive design, and mobile optimization.

### Phase 7: Testing & Refinement (Week 3-4)

Comprehensive testing, data integrity validation, bug fixes, and performance optimization.

### Phase 8: Documentation & Deployment (Week 4)

Complete documentation, deployment preparation, and production readiness assessment.

## Technical Challenges & Solutions

| Challenge | Solution Implemented | Outcome |
| --- | --- | --- |
| **Data Integrity Issues** Orphaned records across tables breaking | Implemented strict foreign key constraints, created automated verification scripts, redesigned | 100% data consistency achieved, |

| Challenge | Solution Implemented | Outcome |
|---|---|---|
| foreign key relationships | data generation to use actual database IDs | zero orphaned records |
| **Authentication Failures** Security answers not matching despite correct input | Added comprehensive debug logging, implemented case-insensitive comparison with whitespace trimming, simplified answer format | Reliable authentication with 100% success rate for correct credentials |
| **Voice Input Restrictions** Browser microphone access blocked without HTTPS | Implemented HTTPS deployment via Gradio share links and Hugging Face Spaces platform | Full voice functionality enabled across all browsers |
| **Session Management** Risk of cross-user data leakage in multi-user scenarios | Implemented unique session IDs with isolated state management per user, strict customer_id filtering | Secure multi-user support with complete data isolation |
| **API Cost Control** Potential for runaway OpenAI API costs | Implemented rate limiting per session, hard usage caps, and monitoring dashboard integration | Predictable costs at $0.01-0.05 per conversation |
| **Context Management** AI losing conversation context across multiple queries | Implemented conversation history tracking with proper message formatting and context window management | Natural multi-turn conversations with maintained context |

# Use Cases & Example Interactions

**Example 1: Complete Authentication Flow**
**User:** "Hello"
**Assistant:** "Welcome to our AI Banking Assistant! To get started, please provide your CX code for authentication."
**User:** "A1234XYZ"
**Assistant:** "Thank you! For security purposes, please answer this question: What is your mother's maiden name?"
**User:** "Smith"
**Assistant:** "Welcome back, John Doe! ✅ Your Account Summary: Customer ID: 1, Checking Account (ID: 1) - Balance: $5,432.18, Savings Account (ID: 2) - Balance: $23,456.89. How can I help you today?"

**Example 2: Balance Inquiry**
**User:** "What's my checking account balance?"
**System Action:** Calls `check_balance(customer_id=1, account_type='checking')`
**Assistant:** "Your checking account balance is $5,432.18. Your account is active and in good standing."

**Example 3: Transaction History**
**User:** "Show me my last 5 transactions"
**System Action:** Calls `get_recent_transactions(customer_id=1, limit=5)`
**Assistant:** "Here are your last 5 transactions:

- Dec 15, 2024 - Walmart - $87.42 (debit)
- Dec 14, 2024 - Salary Deposit - $2,500.00 (credit)
- Dec 13, 2024 - Shell Gas - $45.32 (debit)
- Dec 12, 2024 - Amazon - $156.78 (debit)
- Dec 11, 2024 - Starbucks - $8.45 (debit)"

**Example 4: Advanced Transaction Search**
**User:** "Show me all transactions over $100 in the last week"
**System Action:** Calls `search_transactions(customer_id=1, min_amount=100, days=7)`
**Assistant:** "I found 3 transactions over $100 in the last 7 days:

- Dec 14 - Salary Deposit: $2,500.00
- Dec 12 - Amazon: $156.78
- Dec 10 - Rent Payment: $1,200.00

Total spent: $1,356.78, Total received: $2,500.00"

**Example 5: Loan Eligibility Check**
**User:** "Am I eligible for a $20,000 personal loan?"
**System Action:** Calls `check_loan_eligibility(customer_id=1, loan_amount=20000, loan_type='personal')`
**Assistant:** "Great news! Based on your credit score of 720 and annual income of $65,000, you are eligible for a $20,000 personal loan. Here are your estimated terms:

- Estimated Interest Rate: 6.5%
- Estimated Monthly Payment: $391.32
- Loan Term: 60 months

Would you like more information about our loan products?"

**Example 6: Multi-Account Overview**
**User:** "What are all my accounts?"
**System Action:** Calls `get_all_accounts(customer_id=1)`
**Assistant:** "You have 2 active accounts:

- Checking Account (ID: 1): $5,432.18
- Savings Account (ID: 2): $23,456.89

Total Balance: $28,889.07. All accounts are active and in good standing."

# Future Enhancements

### 🔔 Real-time Notifications

Push alerts for transactions, low balances, unusual activity, and account updates via email or SMS

### 💳 Bill Payment Integration

Direct bill payment functionality through the assistant with payee management and scheduling

### 📊 Financial Insights

AI-powered spending analysis, budget recommendations, and personalized financial advice

### 🌍 Multi-language Support

Expand to Spanish, French, Chinese, and other languages for global reach

### 📱 Mobile Applications

Native iOS and Android apps with push notifications and biometric authentication

### 🤝 Third-party Integration

Connect with Plaid, Stripe, or other financial APIs for real bank data integration

### 📈 Advanced Analytics

Spending trends, category analysis, and predictive budgeting features

### 🔐 Biometric Auth

Fingerprint and face recognition for mobile devices with enhanced security

# Deployment Options

## Option 1: Hugging Face Spaces (Recommended)

**Advantages:**

- Free permanent hosting with no time limits
- Automatic HTTPS with full microphone support
- Professional URL for sharing and portfolio
- Built-in version control with Git integration
- Community visibility and engagement
- Simple deployment process (15 minutes)

**Setup Time:** 15 minutes
**Cost:** Free (pay only for OpenAI API usage ~$0.01-0.05 per conversation)
**URL Format:** https://huggingface.co/spaces/username/ai-banking-assistant

## Option 2: Gradio Share Link

**Quick Testing Solution:**

- Instant deployment with one line of code: `app.launch(share=True)`
- Automatic HTTPS link generation
- Perfect for demos and testing
- No configuration required

**Limitations:**

- Link expires after 72 hours
- Requires your computer to stay running
- Limited bandwidth based on your connection

## Option 3: Cloud Platforms

The application is compatible with major cloud platforms:

✓ **AWS:** EC2, Lambda, or Elastic Beanstalk deployment

✓ **Google Cloud:** App Engine or Cloud Run

✓ **Azure:** App Service or Container Instances

✓ **Railway:** Simple deployment with automatic HTTPS

✓ **Render:** Free tier available with easy deployment

# Project Metrics & Statistics

| 2,000+ | 8 | 5 | 100% |
|---|---|---|---|
| Lines of Code | Core Functions | Database Tables | Test Coverage |

## Code Quality Metrics

- ✓ **Modular Architecture:** Clean separation of concerns across 8+ modules
- ✓ **Error Handling:** Comprehensive try-catch blocks with graceful degradation
- ✓ **Documentation:** Inline comments, docstrings, and comprehensive README
- ✓ **Type Safety:** Type hints for all function parameters and returns
- ✓ **Testing:** Automated verification scripts for data integrity
- ✓ **Security:** Parameterized queries throughout, no SQL injection vulnerabilities
- ✓ **Performance:** Optimized queries with sub-second response times
- ✓ **Maintainability:** Clear code structure with consistent naming conventions

## Technical Specifications

| Metric | Value |
|---|---|
| Python Version | 3.8+ |
| Database Size | ~50 MB with 1000+ transactions |
| API Response Time | < 2 seconds average |

| | |
|---|---|
| Voice Processing | 1-3 seconds total (STT + TTS) |
| Concurrent Users | Unlimited (stateless sessions) |
| Deployment Time | < 15 minutes to production |

## Key Learning Outcomes

### AI/ML Integration

**Skills Gained:**

- OpenAI API mastery
- Function calling implementation
- Prompt engineering
- Context management
- AI tool orchestration

### Database Design

**Skills Gained:**

- Relational database modeling
- Foreign key constraints
- Data integrity validation
- Query optimization
- Transaction management

### Security Practices

**Skills Gained:**

- Two-factor authentication
- Session management
- SQL injection prevention
- Secure coding practices
- Data access control

### Voice Technology

**Skills Gained:**

- Speech recognition (Whisper)
- Voice synthesis (TTS)
- Audio processing
- Multi-modal interfaces
- Accessibility implementation

### Full-Stack Development

**Skills Gained:**

- End-to-end app development
- Backend API design
- Frontend UI/UX
- State management
- System architecture

### Cloud Deployment

**Skills Gained:**

- Production deployment
- Environment management
- Monitoring & logging
- Scaling strategies
- DevOps practices

# Conclusion

The AI Banking Assistant demonstrates the practical application of modern AI technologies in the financial services sector. By combining natural language processing, voice recognition, and secure authentication, the system provides an intuitive and accessible banking experience that showcases the potential for AI to transform customer service across industries.

This project successfully addresses real-world challenges in banking automation while maintaining security, reliability, and user experience as core priorities. The implementation demonstrates proficiency across the full technology stack, from database design to AI integration, culminating in a production-ready application that can serve as a foundation for commercial banking solutions.

### 🎯 Project Impact & Achievements

**Technical Achievement:**

Successfully integrated multiple complex technologies—OpenAI GPT-4o-mini, Whisper API, TTS-1, SQLite, and Gradio—into a cohesive, production-ready application with robust error handling and security measures.

**Business Value:**

Demonstrated how AI can streamline banking operations, reduce customer service costs, improve accessibility through voice interfaces, and provide 24/7 automated assistance without compromising security.

**Innovation:**

Combined voice interaction with secure two-factor authentication to create an accessible yet secure banking experience, addressing both usability and security concerns simultaneously.

**Scalability:**

Built with a modular, scalable architecture ready for horizontal scaling, cloud deployment, and integration with real banking systems through minimal modifications.

### 📚 Technologies Mastered

**AI & Machine Learning:**

- OpenAI GPT-4o-mini
- Function Calling
- Whisper API
- TTS-1 API

**Backend Technologies:**

- Python 3.8+
- SQLite Database
- RESTful APIs
- Session Management

**Frontend & UI:**

- Gradio Framework
- Responsive Design
- Audio Processing
- Real-time Updates

**DevOps & Deployment:**

- Hugging Face Spaces
- Environment Variables
- HTTPS Deployment
- Git Version Control

## Resources & Documentation

| Resource Type | Description | Access |
|---|---|---|
| **Live Demo** | Interactive demonstration of the AI Banking Assistant | https://huggingface.co/spaces/[username]/ai-banking-assistant |
| **Source Code** | Complete codebase with all modules and tools | GitHub Repository: [Your URL] |
| **Setup Guide** | Detailed installation and configuration instructions | README.md in repository |

| Resource Type | Description | Access |
|---|---|---|
| **API Documentation** | Function specifications and usage examples | docs/ folder in repository |
| **Demo Video** | Complete walkthrough showing all features | [YouTube/LinkedIn URL] |
| **Database Schema** | ER diagram and table specifications | Included in this document (Page 3) |

### External Documentation

- ✓ **OpenAI API Documentation:** https://platform.openai.com/docs
- ✓ **Gradio Documentation:** https://www.gradio.app/docs
- ✓ **Hugging Face Spaces Guide:** https://huggingface.co/docs/hub/spaces
- ✓ **Python SQLite Tutorial:** https://docs.python.org/3/library/sqlite3.html

## Contact & Collaboration

**Interested in this project or similar AI applications?**

I'm open to:

- Collaboration on AI/ML projects
- Technical discussions and knowledge sharing
- Opportunities in AI engineering and fintech
- Speaking engagements about AI implementation
- Consulting on AI integration strategies

**Connect with me:**

LinkedIn: https://www.linkedin.com/in/micheal-olaniyi/
GitHub: https://github.com/Purepen
Email: https//www.purepen1376@gmail.com
Youtube: https://youtu.be/Hvwg5XZ3TR0

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## AI Banking Assistan

A demonstration of modern AI technology in financial services

**Built with:** OpenAI GPT-4o-mini | Python | Gradio | SQLite | Whisper API | TTS-1 API

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By Micheal Olaniyi [AI/Machine Learning Engineer]